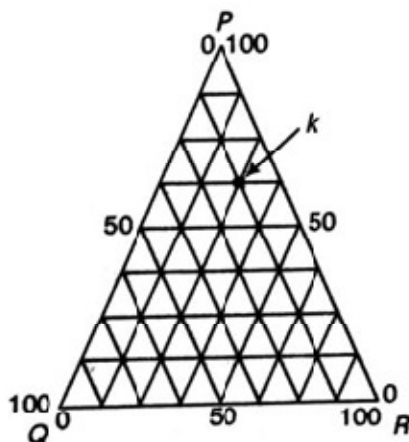


B.TECH
(SEM VI) THEORY EXAMINATION 2022-23
MASS TRANSFER –II

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 10 = 20**

- (a) What is Relative volatility? Give expression for a binary mixture?
- (b) Write down the assumptions of MacCabe Thiele method.
- (c) What is the use of entrainer in azeotropic distillation?
- (d) Write usefulness of Poncho-Savarit method?
- (e) For a system containing species P, Q and R, composition at point k on the ternary plot is



- (f) Write the importance of binodal curve in phase diagram for a biphasic system in liquid liquid extraction.
- (g) Define: plait point.
- (h) What is the principle of leaching?
- (i) State the difference between adsorption and extraction.
- (j) Give two examples for leaching operation.

SECTION B**2. Attempt any three of the following:****10 x 3 = 30**

- (a) Discuss briefly the simple distillation process and state its applications.
- (b) A feed containing 50mol% benzene and 50 mol% toluene is to be distilled in a fractionating column to produce a distillate containing 90 mol% benzene and bottoms containing 90 mol% toluene. Feed rate to the column is 10000 kg/h and feed is at its bubble point. The operating reflux ratio is 1.5 times the minimum and the overall plate efficiency is 0.75. Determine the actual number of plates required. The relative volatility is 2.28.
- (c) Write briefly about selection of solvent for liquid liquid extraction.
- (d) 1000 kgs/hr of a dioxane-water mixture containing 25% dioxane (by weight) is to be continuously extracted in counter fashion with benzene to remove 95% of the dioxane. Assume benzene and water are immiscible with each other. The equilibrium data are as follows:

Wt % dioxane in water phase	5.1	18.9	25.2
Wt % dioxane in benzene phase	5.2	22.5	32.0

- (i) Calculate the minimum solvent required in kg/hr and (ii) If 900 kg/hr of solvent is used; calculate the number of theoretical stages required.
- (e) Discuss Steam distillation with neat diagram. Also give application of steam distillation.

SECTION C

3. Attempt any *one* part of the following: **10 x 1 = 10**
- Draw a continuous distillation column. Indicate the parts and sections in the continuous distillation column. Explain the constructional features, working principle and application of it.
 - Explains the following terms
(a) Reflux ratio (b) Infinite or total reflux ratio (c) Minimum reflux ratio (d) Optimum reflux ratio
4. Attempt any *one* part of the following: **10 x 1 = 10**
- Tabulate different conditions of feed along with their q values (or) formula. Represent q lines for different feed conditions on the equilibrium diagram.
 - A mixture containing benzene and toluene with 40% benzene and 60% toluene is to be separated in a fractionating column to give product containing 96% benzene and bottom product containing 95% toluene. Feed is a mixture of two-third vapor and one-third liquid. Find the number of theoretical stages required if the reflux ratio of 1.5 times the minimum is used. (Relative volatility = 2.5)
5. Attempt any *one* part of the following: **10 x 1 = 10**
- Explain the following term:
i) Selectivity ii) Distribution coefficient iii) Tie Line iv) Operating Line
 - If 100 kg of a solution of acetic acid (C) and water (A) containing 30% acid is to be extracted three times with isopropyl ether (B) at 20°C, using 40 kg of solvent in each stage, determine the quantities and compositions of the various streams. How much solvent would be required if the same final raffinate concentration were to be obtained with one stage?

Water Layer			Isopropyl ether layer		
Wt% Acetic acid 100x	Water	Isopropyl ether	Acetic acid, 100 y*	Water	Isopropyl ether
0.69	98.1	1.2	0.18	0.5	99.3
1.41	97.1	1.5	0.37	0.7	98.9
2.89	95.5	1.6	0.79	0.8	98.4
6.42	91.7	1.9	1.93	1.0	97.1
13.30	84.4	2.3	4.82	1.9	93.3
25.50	71.1	3.4	11.40	3.9	84.7
36.70	58.9	4.4	21.60	6.9	71.5
44.30	45.1	10.6	31.10	10.8	58.1
46.40	37.1	16.5	36.20	15.1	48.7

6. Attempt any *one* part of the following: **10 x 1 = 10**
- Draw multistage counter current leaching and show it on $N-x-y$ plot with relevant equations?
 - Explain in detail about the total material balance for the Single stage and Multistage Counter Current Extraction with neat sketch.

7. Attempt any *one* part of the following:

10 x 1 = 1

- (a) Distinguish between the two types of adsorption phenomena bringing out their essential features
- (b) An aqueous solution is colored by small amounts of impurity which is to be removed by adsorption on activated carbon. The color intensity which is proportional to the concentration of the colored substance was measured on an arbitrary scale. It is desired to reduce the color to 10 % of its original value, 9.6. Estimate the amount of adsorbent used for single stage.

Kg of carbon/ kg of solution	0	0.001	0.004	0.008	0.02	0.04
Equilibrium Color	9.6	8.6	8.3	4.3	1.7	0.7

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